CALVIN BANDOL SANDALS: RECYCLED PRODUCT INNOVATION UTILIZING WASTE TIRES AND EVA (ETHYLENE VINYL ACETATE) FOAM TO REDUCE UNUSED OBJECTS AND INCREASE THE PRODUCTIVITY OF THE SUSTAINABLE CREATIVE ECONOMY IN PURWOKERTO

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A sandal is often considered a trivial item, even though it keeps a long chain of material. Behind the soles we step on are tires that once traveled thousands of kilometers, and EVA (ethylene vinyl acetate) foam, known locally as *busa ati*, cut from larger sheets. After becoming leftovers, both turn into unused objects that accumulate, throw away, or burn. But in Purwokerto, the creative hands of the artisans are able to create a new solution, namely by reprocessing them into everyday items, namely bandol sandal products.

The scale of the problem is real. According to the National Waste Management Information System (SIPSN), national waste generation in 2025 was approximately 34.6 million tons, of which 68.5 percent had not been managed.[1] A 2025 study estimates that Greater Jakarta (Jabodetabek) and West Java alone produce 2.5–3 million used tires per year, of which only about 20 percent are optimally utilized.[2] From the footwear industry, EVA commonly used for midsoles and outsoles produces leftover production that generally ends up in landfills, while a truly established EVA recycling process is not yet available.[3] EVA sponge waste is also difficult to decompose and takes a long time, especially since the practice of disposing and burning it pollutes the environment.[4]

Regulatively, waste is declared managed if it enters facilities such as waste banks, recycling centers, or integrated waste processing sites.[5] Outside this formal channel, recycling by small businesses often runs on their own, is poorly recorded, and tends to operate independently as a community initiative, as in the case of craft products such as bandol sandals. As for the meaning of the name, “bandol” is a

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[1] Ministry of Environment, "*Indicative Data – National Waste Management Information System (SIPSN)*" (https://sampahnasional.kemenlh.go.id/portal-indikatif/, Accessed on October 27, 2026).

[2] Ono Suparno, Suprihatin, Nastiti Siswi Indrasti, Hartrisari Hardjomidjojo, Titi Candra Sunarti, Prasetyo Hadi Utomo, Efri Yulistika, Yunus Triyonggo, & Delfania Matasik, "*Used Tire Ecosystem for the Reclaimed Rubber Industry Development at PT Bridgestone Tire Indonesia*" Vol. 35, No. 2, August 2025, p. 208–216.

[3] Iacopo Bianchi ,Archimede Forcellese ,Michela Simoncini, & Alessio Vita, “*Mechanical Characterization and Sustainability Assessment of Recycled EVA for Footwear*” 126, no. 7–8 (2023): 3149–3160.

[4] Ovy Chandra Singhasari, Imma Ma'rifatus Sayidah Ummu, Tiara Daffa Nur Aini, Indari Indari, Musthika Wida Mashitah, & Alfunnafi' Fahrul Rizzal, "*MONTESOL: Efforts to Improve the Economy through the use of Sandal Rubber Waste as Educational Game Tools*" 9, No. 6 (2025).

[5] Ministry of Environment and Forestry, "*Regulation of the Minister of Environment and Forestry Number 6 of 2022 concerning the National Waste Management Information System"* (https://jdih.kehutanan.go.id/new2/home/portfolioDetails3/2022pmlhk006_menlhk_04112022103639.pdf/6/2022/4/1104, Accessed on October 27, 2026).

contraction of "ban bodol," meaning an unused tire. This craft has developed within the community since around the 1950s and is centered in Banaran (Kebanaran), Pasir Kidul Subdistrict, West Purwokerto.[6]

The Calvin Bandol Sandals production house is one of many sites that produce recycled crafts in the form of bandol sandals. Mr. Justo, one of the artisans, explained that this production house can produce at least 100 kodi (one kodi equals 20 pieces), or about 2,000 pairs of bandol sandals, per month, priced at Rp200,000 to Rp250,000 per kodi. The raw materials are dominated by used tires, EVA foam, glue, thread, and several other minor materials, and some of them, such as the tires and EVA foam, are brought in from outside the region, for example from Greater Jakarta. Then, the tire and foam sheets are cut according to the sole pattern, glued in layers, and then flattened with a screw-pressed roller machine. The edges of the soles are smoothed with a grinding machine, the straps are fitted with a sewing machine, and the sandals are packaged in plastic per bundle for sale. In the business practice of the creative industry (UMKM), the making of bandol sandals has, in general, solved at least three problems. First, the raw material problem: used tires, which some people regard as waste, become a cheap and wear-resistant source of material, so that an environmental burden is turned into economic value.[7] Second, the problem of livelihood: this small industry becomes a source of income for local workers to meet their daily needs.[8] Third, the problem of affordable, quality products: in the early days, people needed footwear that was affordable yet highly durable, a need met by bandol sandals, which are very strong, abrasion-resistant, tough, and pocket-friendly.

However, behind that, there is still a continuous problem that still arises at the end of the product creation phase, namely the residual derivative waste during the process of making bandol sandals which has previously been sorted from the best parts. This residue consists of tire pieces, EVA foam, sewing thread, and

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[6] Budi Setyawan, "*Sandal Bandol Center, Kebanaran Village, Pasir Kidul, Purwokerto: Reaping Blessings from Tire Waste*" (https://www.suaramerdeka.com/ekonomi/0910411946/di-sentra-kerajinan-bandol-ban-bekas-bisa-disulap-jadi-kerajinan-ini-lho, Accessed on October 27, 2026).

[7] Ibid.

[8] Permata Putra Sejati, "*The Success Story of Banyumas Bandol Sandal Artisan Zaenal Abidin, Turning Used Tires into Classy Goods*" (https://banyumas.tribunnews.com/2024/12/07/kisah-sukses-perajin-sandal-bandol-khas-banyumas-zaenal-abidin-ubah-ban-bekas-jadi-barang-berkelas, Accessed on October 27, 2026).

several other materials that are set aside. On this basis, it can be concluded that the process of reprocessing waste can itself give rise to new waste, for which solutions need more attention. Waste banks and other efforts, whether government support or environmental movements, are needed to better support the sustainability of community creativity in tackling environmental problems.



Figure 1 The process of making bandol sandals

The description of this waste-based creative industry still needs to be read critically. This study of the village promotion design concluded that Kampung Bandol is still poorly known to the surrounding community because it does not have characteristics and lacks promotion,[9] and local traders say that buyers are increasingly unfamiliar with bandol sandals because they are more accustomed to factory-made sandals.[10] In other words, material innovation has been successful, but market innovation is still lagging behind. There is a need for education and teaching in the implementation of a branding system that is able to raise the identity

[9] Syahgita Putri Aura, "*Designing the Mascot of Kampung Bandol Kebanaran, West Purwokerto District*" (https://repository.ittelkom-pwt.ac.id/10731/3/ABSTRAK%20Indo.pdf, Accessed on October 27, 2026).

[10] Sekar Pramudya Damayanti, "*Faded Traces: The Charm of Banyumas Bandol Sandals That Are Almost Forgotten*" (https://kumparan.com/sekarpramudya-damayanti/jejak-yang-pudar-pesona-sandal-bandol-khas-banyumas-yang-hampir-terlupakan-21wpRFeoAT6, Accessed on October 26, 2026).

and speciality of bandol sandal products as a regional icon that has competitiveness like premium products outside. Such as teaching about knowledge about the importance of the function of *brand identity* in attracting consumer interest in the market era that has shifted to the digital world, especially marketing in online buying and selling platforms and social media promotion.

The relevance of its recycling value lies in the way in which the product extends the life of the material. Tire rubber that originally ended up as waste is diverted into soles that are reused daily, while the foam left over from the footwear industry is given a new function as a sole layer. This pattern is in line with the circular economy principle, which is to keep materials circulating for as long as possible and reduce the need for new raw materials. The global context reinforces this relevance, where flip-flops only have a lifespan of about two years; around three billion pairs are produced annually, and some reports say that around 200 million pairs are thrown away.[11] One example is the Ocean Sole organization in Kenya, which recycles discarded sandals into sculptures while creating livelihoods for local artisans. Bandol sandals follow the same logic but yield functional items for daily use, so they do not depend on the art market.

In terms of sustainability, the three pillars need to be looked at together. Economically, this business lasts for decades, but it depends on the supply of tires from outside the region and a narrow market. Socially, it absorbs local workers and becomes a regional identity that is now strengthened through training.[12] In terms of the environment and work safety, the remaining pieces that accumulate are still a record, and the aspect of work safety is also worth considering because there are still many workers who do not use safety footwear and gloves when using machines and tools that are prone to injury.

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[11] Dinda Rahmania, "*Looking at How Ocean Sole Turns Used Sandal Waste into a Work of Art*" (https://greennetwork.id/gna-knowledge-hub/menengok-bagaimana-ocean-sole-mengubah-sampah-sandal-bekas-menjadi-karya-seni/, Accessed on October 30, 2026).

[12] Diva Nurafriani Shafa, Rico Ahmad Dzulqorni, & Arif Andri Wibowo, "*Economic Development of the 'Bandol' Area of Banaran Village, Pasir Kidul Village, Purwokerto*" Vol. 3, No. 1 (2024) p. 12–20.



Figure 2 Manufacturing process and residual derivative waste

From a policy (government) perspective, the authors see a recognition gap, because waste is only counted as managed once it enters certain facilities (e.g., waste banks).[13] Used tires absorbed by bandol sandal artisans are not recorded as a waste management achievement, even though physically they are kept from piling up or being burned, since most of them have been recycled into UMKM products. A thorough record of the volume of material absorbed is needed from the local environmental agency, for example by making production houses or artisan associations data partners, which would make this contribution visible and open up opportunities for mentoring. Several steps can strengthen its sustainability: establishing partnerships with tire collectors and the local footwear industry so that supply connections are stable and closer; processing debris into secondary products such as pot mats or keychains; implementing simple occupational safety standards; and strengthening brands and online sales based on stories "from used tires to sandals" so that the value of recycling is known to buyers.

[13] Ministry of Environment and Forestry, op. cit.



Figure 3 Sandal bandol products

Bandol sandals show that recycling does not always require a large-scale facility; it can grow from the skills of artisans and the needs of the market on a daily basis. This innovation solves the problems of raw materials, livelihoods, and the need for affordable yet durable footwear, while addressing partial access to employment opportunities within the community, which is in line with the circular economy. Their sustainability now depends on the management of residual production, occupational safety, and market strengthening. With these improvements, the bandol sandals deserve to be positioned not just as nostalgic crafts, but as an example of community-based recycling practices that are relevant to advancing Indonesia’s eco-friendly creative industry sector.

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